

Work Breakdown Structure (WBS)

Hyperlocal Courier Dispatch & Tracking Engine

Problem Statement #23

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The project is broken down into seven work areas. Each implementation package is tied to the requirement it delivers and to the Jira epic it belongs to.

WBS ID	Work package	Related requirement	Jira epic	Status
1	Project Management			
1.1	Create GitHub repository and README	-	-	Done
1.2	Set up Jira Kanban, Scrum and Bug Tracking projects	-	-	Done
1.3	Sprint planning and story point estimation	All	All epics	Done
2	Requirements Engineering			
2.1	Functional and non-functional requirements table	FR-001 to FR-005, NFR-001, NFR-002	-	Done
2.2	UML use-case diagram and use-case flow specification	All	-	Done
2.3	Requirements Traceability Matrix	All	-	Done
2.4	Software Requirements Specification and WBS	All	-	Done
3	Design			
3.1	System architecture design	All	All epics	Planned
3.2	Module and interface design (sender, rider, dispatch engine, Routing / GPS interface)	All	All epics	Planned
4	Implementation			
4.1	Courier request creation module	FR-002	Courier Request and Dispatch	Planned
4.2	Nearest-rider matching within 3 km	FR-001	Courier Request and Dispatch	Planned
4.3	Rider accept / reject handling and reassignment	FR-003	Courier Request and Dispatch	Planned
4.4	Route generation and multi-stop optimization	FR-004	Routing and Live Tracking	Planned
4.5	Live GPS and status updates to the sender (under 2 s)	NFR-001	Routing and Live Tracking	Planned
4.6	OTP generation, verification and delivery completion	FR-005	Delivery Completion and Reliability	Planned
5	Testing			
5.1	Test planning and test case design from acceptance criteria	All	All epics	Planned
5.2	Functional testing of FR-001 to FR-005	FR-001 to FR-005	All epics	Planned
5.3	Performance and security test (2-second latency under peak load)	NFR-001	Routing and Live Tracking	Planned

WBS ID	Work package	Related requirement	Jira epic	Status
5.4	Availability monitoring (99.5%)	NFR-002	Delivery Completion and Reliability	Planned
5.5	Bug fixing and retesting (Bug Tracking project)	All	-	Planned
6	Deployment			
6.1	Environment and infrastructure setup	NFR-002	Delivery Completion and Reliability	Planned
6.2	Deployment and release	All	-	Planned
7	Documentation and Submission			
7.1	Final documentation, README and repository submission	All	-	Planned

Status: **Done** = completed in the current labs; **Planned** = to be carried out in later labs.